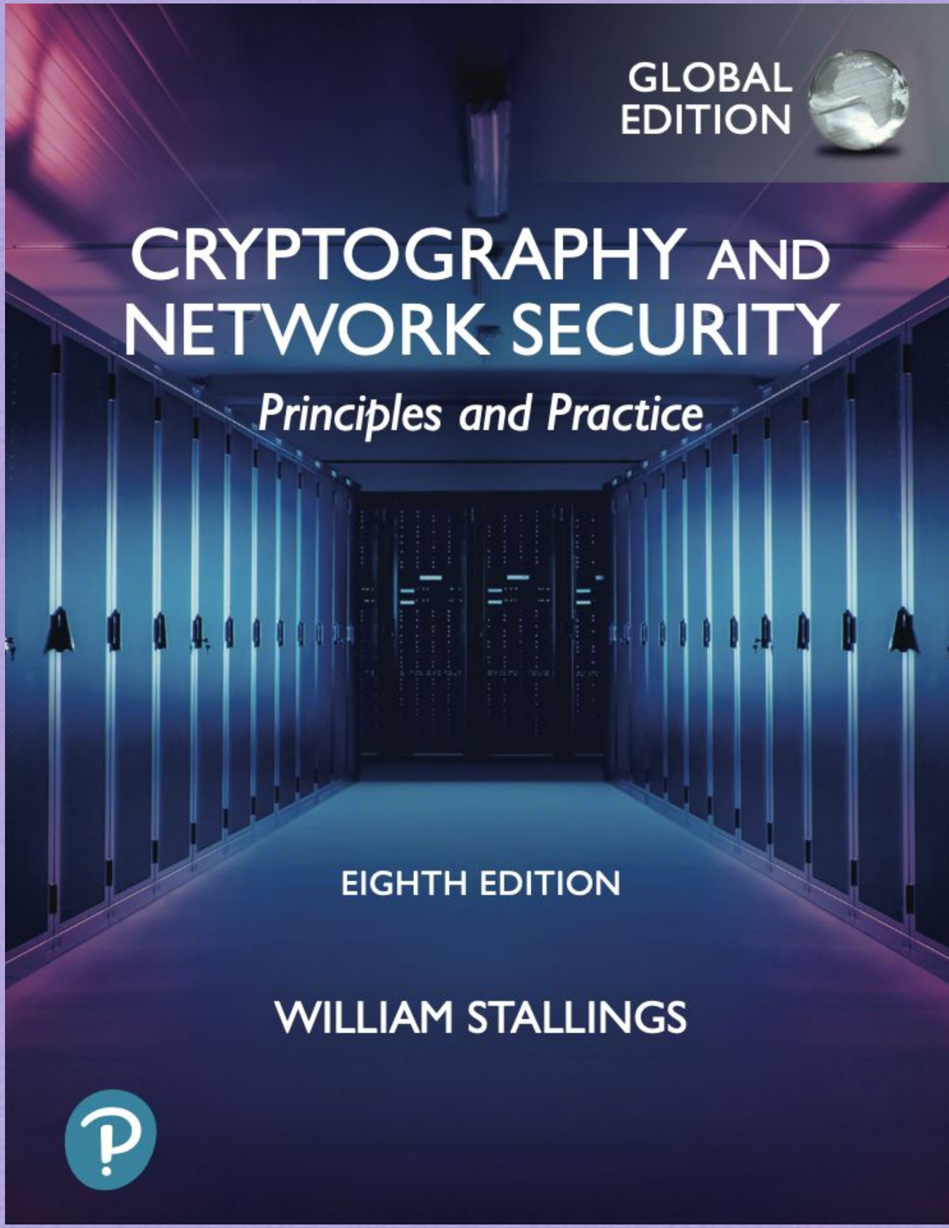
University of Nevada – Reno
Computer Science & Engineering Department

CS454/654 Reliability and Security of Computing Systems - Fall 2024



CHAPTER 10

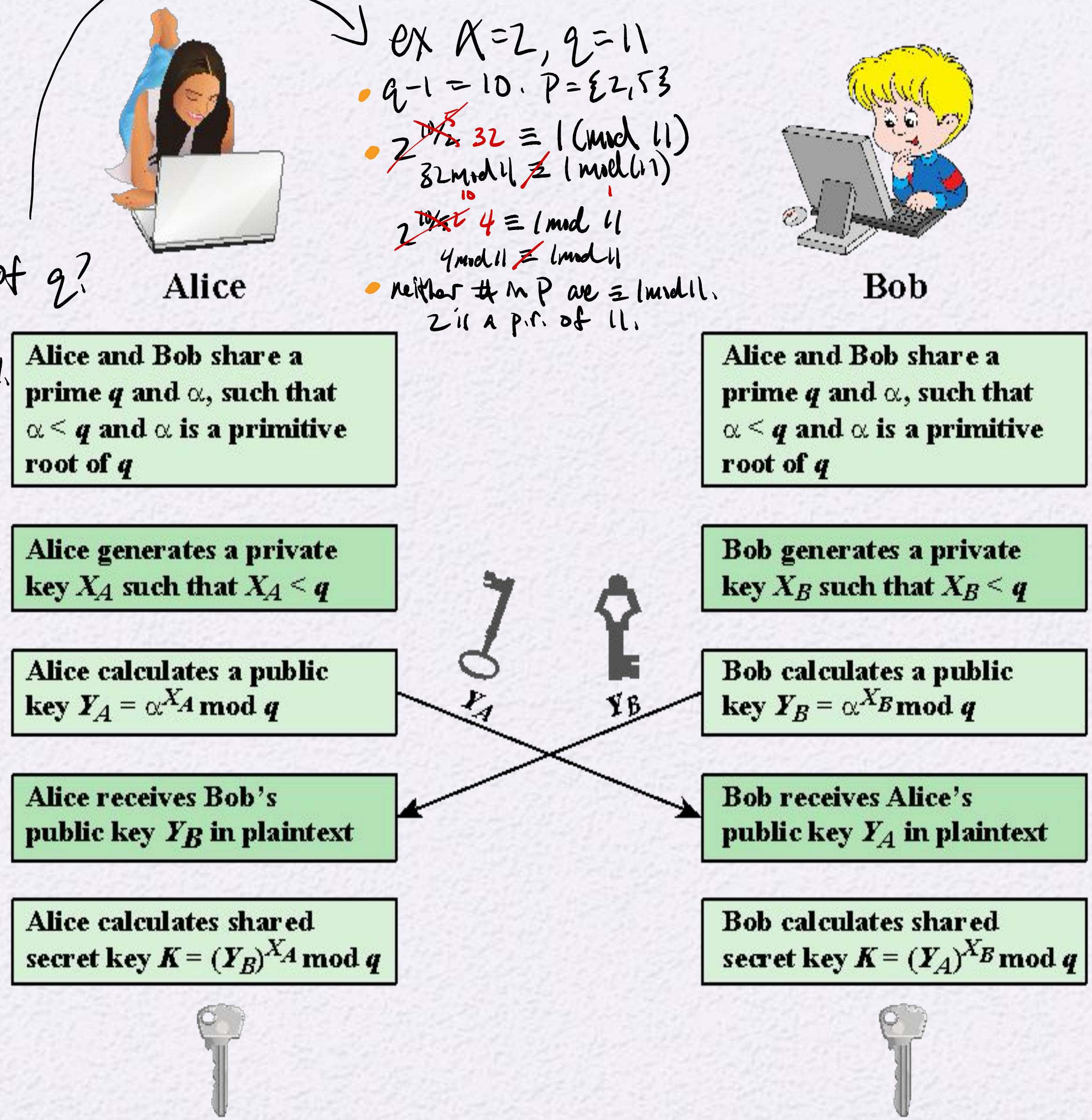
# Other Public-Key Cryptosystems

# Diffie-Hellman Key Exchange

- First published public-key algorithm
- A number of commercial products employ this key exchange technique
- Purpose is to enable two users to securely exchange a key that can then be used for subsequent symmetric encryption of messages
- The algorithm itself is limited to the exchange of secret values
- Its effectiveness depends on the difficulty of computing discrete logarithms

$\alpha$ is a primitive root of $q$?
- order of $q$ is $q-1$.
  Find all prime factors of $q-1$.
- Check if $\alpha^{(q-1)/p} \equiv 1 \bmod q$ for all prime factors $p_n$.
- If $\alpha^{(q-1)/p_n} \not\equiv 1 \bmod q$ for all prime factors $p_n$ of $q-1$, then $\alpha$ is a primitive root of $q$.

ex $\alpha=2$, $q=11$
- $q-1=10$. $P=\{2,5\}$
- $2^{\cancel{10/2}\,5}$ $32 \equiv 1 \pmod{11}$
  $32 \bmod 11 \not\equiv 1 \bmod(11)$
- $2^{\cancel{10/5}\,2}$ $4 \equiv 1 \bmod 11$
  $4 \bmod 11 \not\equiv 1 \bmod 11$
- neither # in P are $\equiv 1 \bmod 11$.
  2 is a p.r. of 11.




difficult to separate $X_A$ from $Y_A$

likewise

**Figure 10.1 Diffie-Hellman Key Exchange**

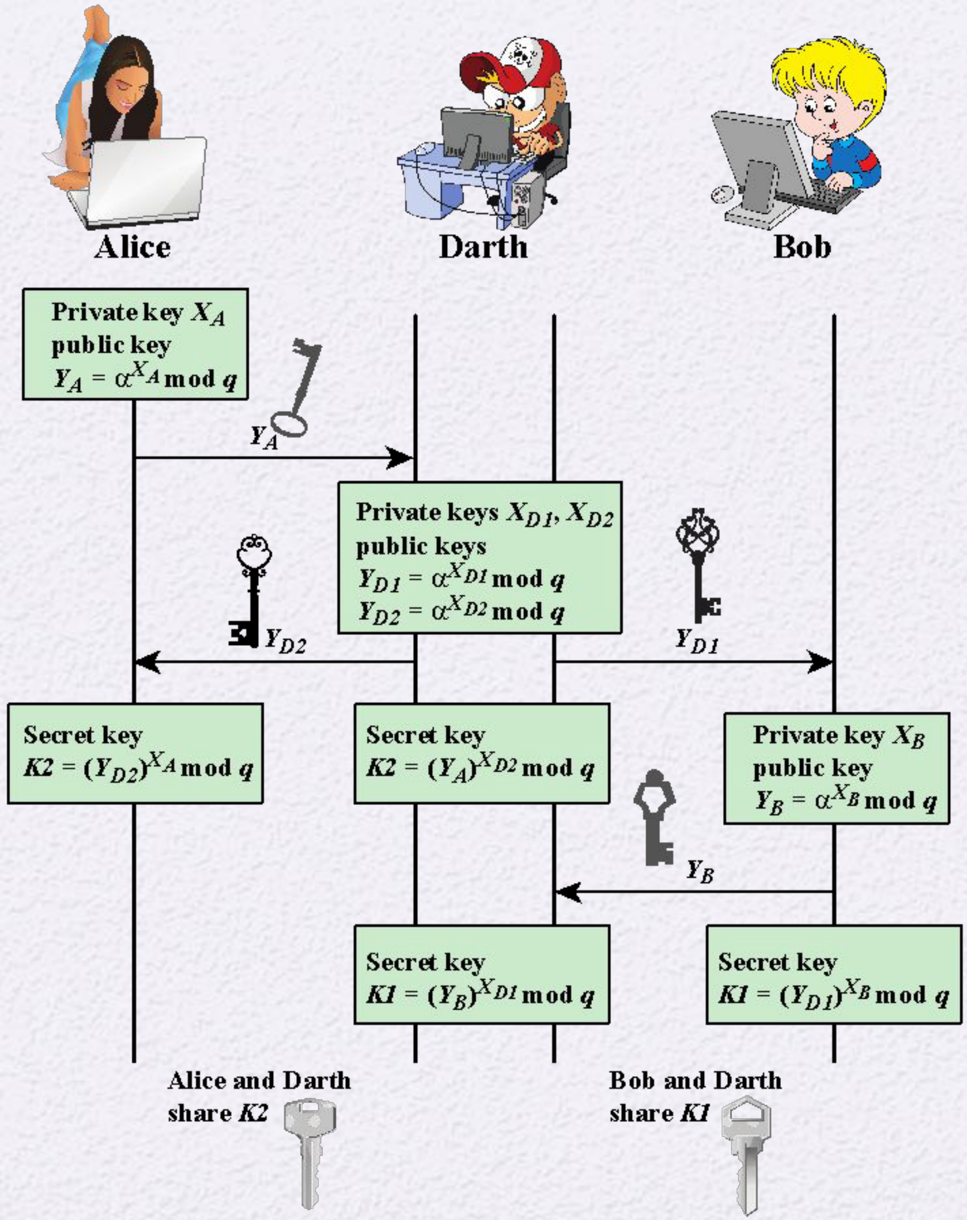



**Figure 10.2 Man-in-the-Middle Attack**

# ElGamal Cryptography

Announced in 1984 by T. Elgamal

Public-key scheme based on discrete logarithms closely related to the Diffie-Hellman technique

Used in the digital signature standard (DSS) and the S/MIME e-mail standard

Global elements are a prime number *q* and *a* which is a primitive root of *q*

Security is based on the difficulty of computing discrete logarithms

refer to slide #4

| Global Public Elements | |
|---|---|
| $q$ | prime number |
| $\alpha$ | $\alpha < q$ and $\alpha$ a primitive root of $q$ |

| Key Generation by Alice | |
|---|---|
| Select private $X_A$ | $X_A < q - 1$ |
| Calculate $Y_A$ | $Y_A = \alpha^{X_A} \bmod q$ |
| Public key | $\{q, \alpha, Y_A\}$ |
| Private key | $X_A$ |

| Encryption by Bob with Alice's Public Key | |
|---|---|
| Plaintext: | $M < q$ |
| Select random integer $k$ | $k < q$ |
| Calculate $K$ | $K = (Y_A)^k \bmod q$ |
| Calculate $C_1$ | $C_1 = \alpha^k \bmod q$ |
| Calculate $C_2$ | $C_2 = KM \bmod q$ |
| Ciphertext: | $(C_1, C_2)$ |

| Decryption by Alice with Alice's Private Key | |
|---|---|
| Ciphertext: | $(C_1, C_2)$ |
| Calculate $K$ | $K = (C_1)^{X_A} \bmod q$ |
| Plaintext: | $M = (C_2 K^{-1}) \bmod q$ |

**Figure 10.3 The ElGamal Cryptosystem**

# Elliptic Curve Arithmetic

- Most of the products and standards that use public-key cryptography for encryption and digital signatures use RSA
  - The key length for secure RSA use has increased over recent years and this has put a heavier processing load on applications using RSA
    default nowadays is 2048-bit
- Elliptic curve cryptography (ECC) is showing up in standardization efforts including the IEEE P1363 Standard for Public-Key Cryptography
  standardization group
- Principal attraction of ECC is that it appears to offer equal security for a far smaller key size

# Abelian Group

- A set of elements with a binary operation, denoted by •, that associates to each ordered pair $(a, b)$ of elements in G an element $(a \cdot b)$ in G, such that the following axioms are obeyed:

like 456 automata!

**(A1) Closure:** If $a$ and $b$ belong to G, then $a \cdot b$ is also in G

**(A2) Associative:** $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c$ in G

**(A3) Identity element:** There is an element $e$ in G such that $a \cdot e = e \cdot a = a$ for all $a$ in G

**(A4) Inverse element:** For each $a$ in G there is an element $a'$ in G such that $a \cdot a' = a' \cdot a = e$

**(A5) Commutative:** $a \cdot b = b \cdot a$ for all $a, b$ in G

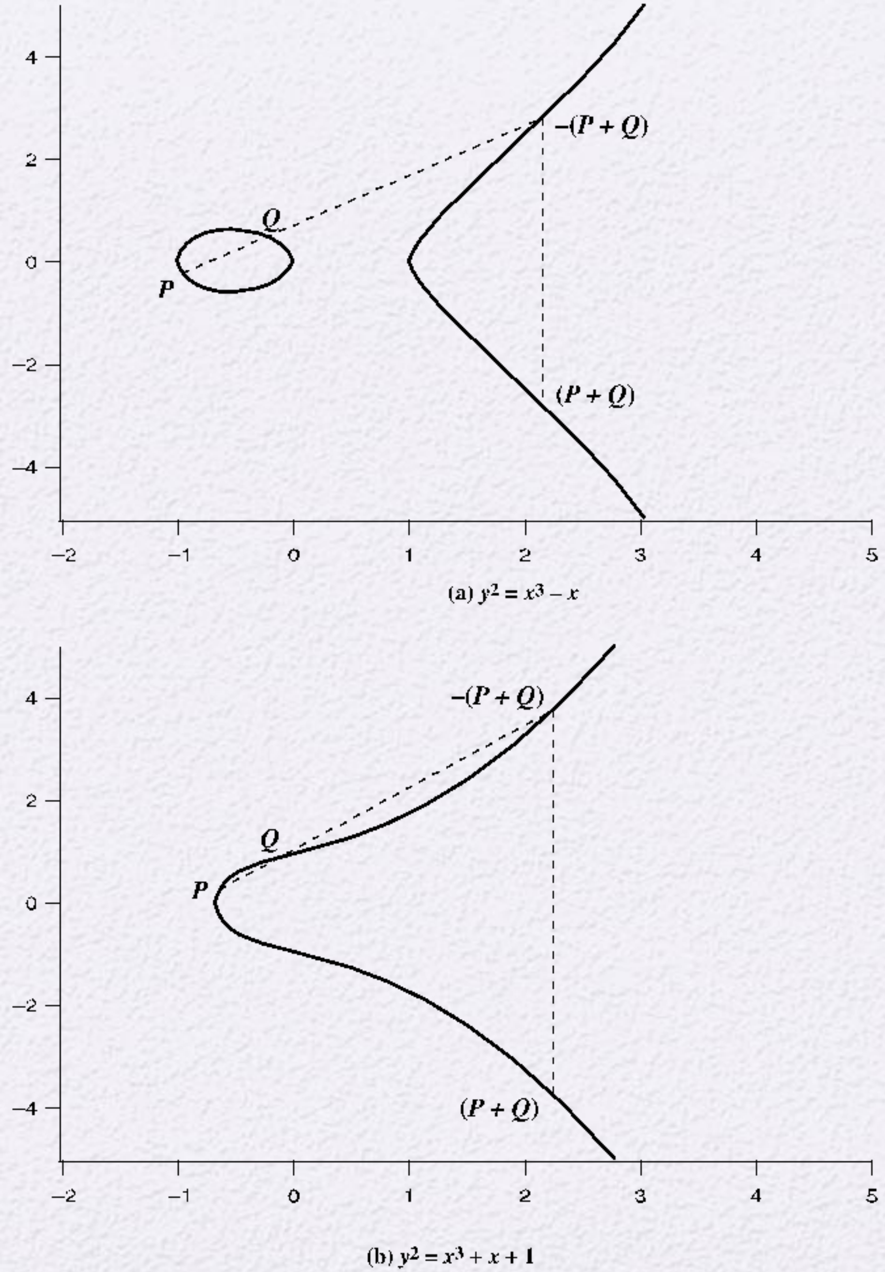
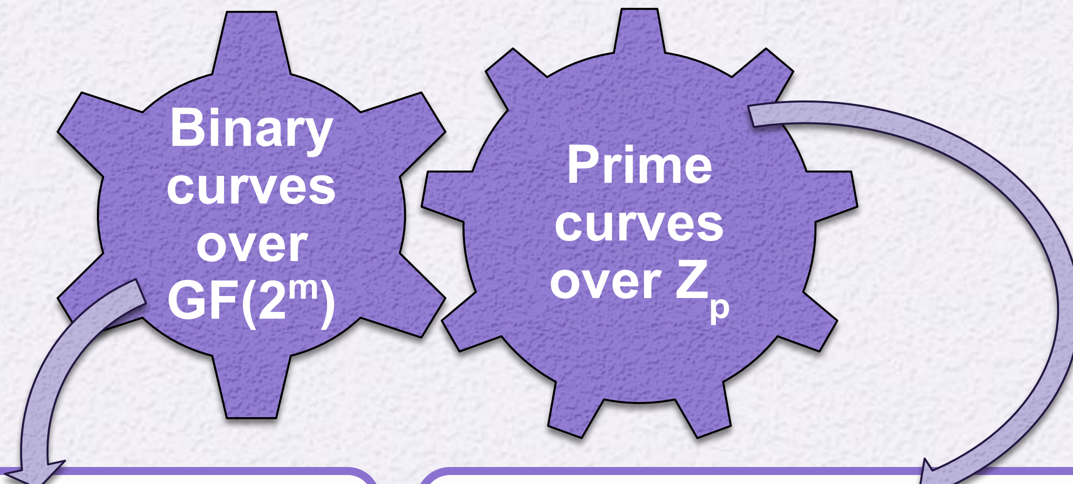



**Figure 10.4 Example of Elliptic Curves**

# Elliptic Curves Over $Z_p$

- Elliptic curve cryptography uses curves whose variables and coefficients are finite
- Two families of elliptic curves are used in cryptographic applications:




- **Variables and coefficients all take on values in GF($2^m$) and in calculations are performed over GF($2^m$)**
- **Best for hardware applications**

- **Use a cubic equation in which the variables and coefficients all take on values in the set of integers from 0 through p-1 and in which calculations are performed modulo p**
- **Best for software applications**

# Table 10.1
## Points (other than O) on the Elliptic Curve $E_{23}(1, 1)$

| | | |
|---|---|---|
| (0, 1) | (6, 4) | (12, 19) |
| (0, 22) | (6, 19) | (13, 7) |
| (1, 7) | (7, 11) | (13, 16) |
| (1, 16) | (7, 12) | (17, 3) |
| (3, 10) | (9, 7) | (17, 20) |
| (3, 13) | (9, 16) | (18, 3) |
| (4, 0) | (11, 3) | (18, 20) |
| (5, 4) | (11, 20) | (19, 5) |
| (5, 19) | (12, 4) | (19, 18) |

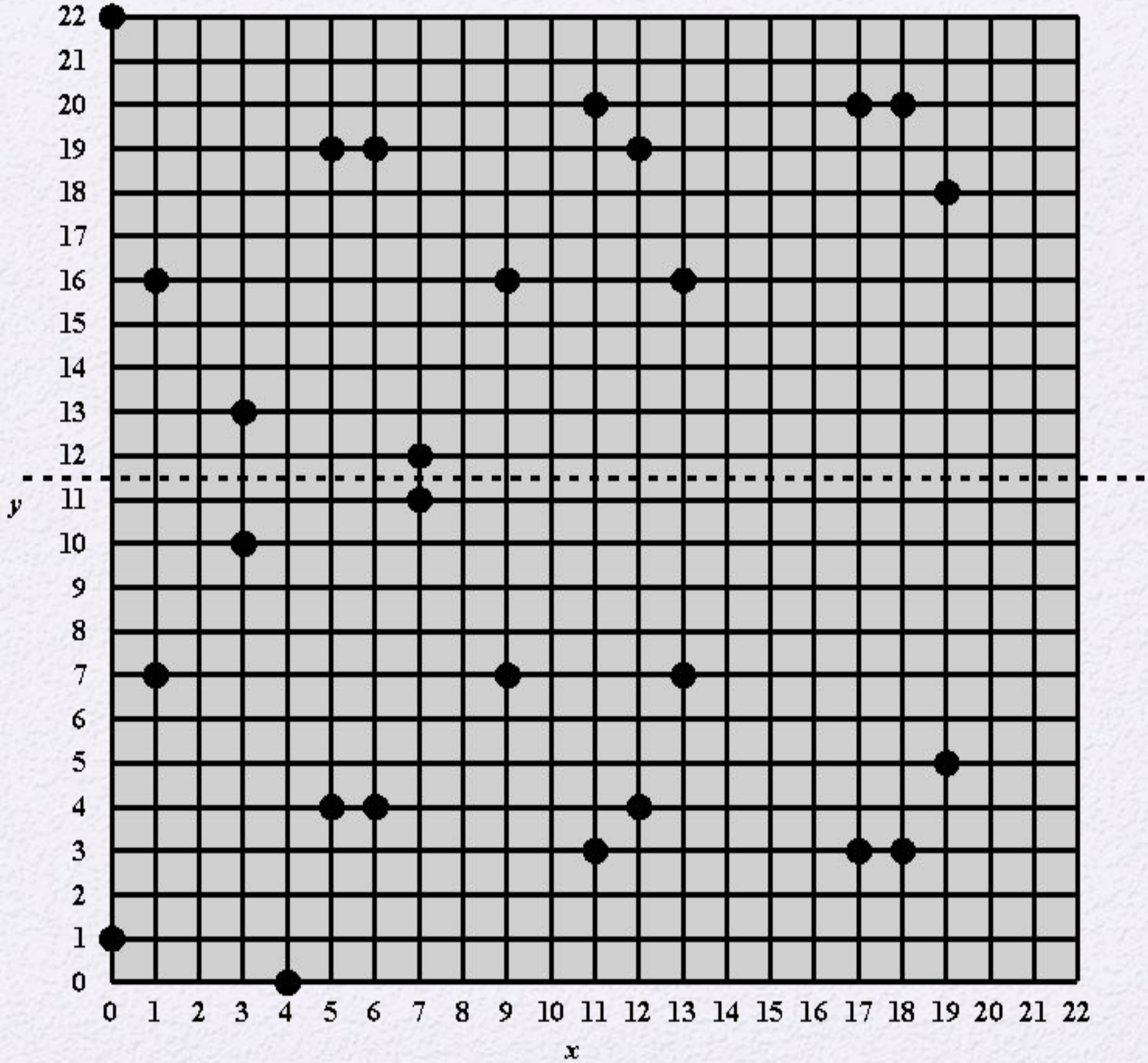



**Figure 10.5 The Elliptic Curve $E_{23}(1,1)$**

# Elliptic Curves Over GF($2^m$)

- Use a cubic equation in which the variables and coefficients all take on values in GF($2^m$) for some number *m*

- Calculations are performed using the rules of arithmetic in GF($2^m$)

- The form of cubic equation appropriate for cryptographic applications for elliptic curves is somewhat different for GF($2^m$) than for $Z_p$
    - It is understood that the variables *x* and *y* and the coefficients *a* and *b* are elements of GF($2^m$) and that calculations are performed in GF($2^m$)

# Table 10.2

## Points (other than O) on the Elliptic Curve $E_{2^4}(g^4, 1)$

| | | |
|---|---|---|
| $(0, 1)$ | $(g^5, g^3)$ | $(g^9, g^{13})$ |
| $(1, g^6)$ | $(g^5, g^{11})$ | $(g^{10}, g)$ |
| $(1, g^{13})$ | $(g^6, g^8)$ | $(g^{10}, g^8)$ |
| $(g^3, g^8)$ | $(g^6, g^{14})$ | $(g^{12}, 0)$ |
| $(g^3, g^{13})$ | $(g^9, g^{10})$ | $(g^{12}, g^{12})$ |

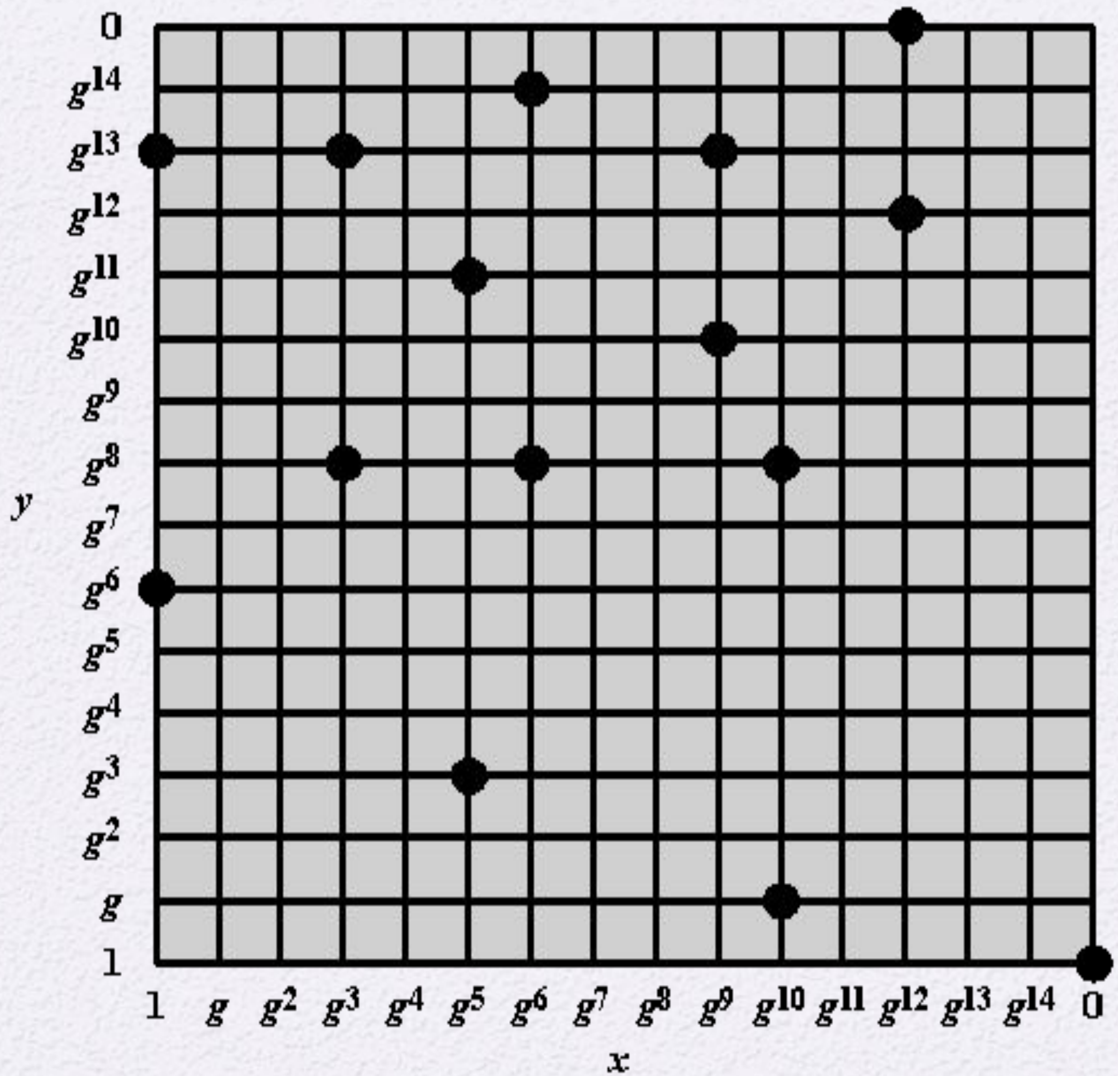



**Figure 10.6 The Elliptic Curve $E_{2^4}(g^4, 1)$**

# Elliptic Curve Cryptography (ECC)

- Addition operation in ECC is the counterpart of modular multiplication in RSA
- Multiple addition is the counterpart of modular exponentiation

To form a cryptographic system using elliptic curves, we need to find a "hard problem" corresponding to factoring the product of two primes or taking the discrete logarithm

- $Q=kP$, where $Q$, $P$ belong to a prime curve
- Is "easy" to compute $Q$ given $k$ and $P$
- But "hard" to find $k$ given $Q$, and $P$
- Known as the elliptic curve logarithm problem

**Global Public Elements**

| | |
|---|---|
| $E_q(a, b)$ | elliptic curve with parameters $a$, $b$, and $q$, where $q$ is a prime or an integer of the form $2^m$ |
| $G$ | point on elliptic curve whose order is large value $n$ |

**User A Key Generation**

| | |
|---|---|
| Select private $n_A$ | $n_A < n$ |
| Calculate public $P_A$ | $P_A = n_A \times G$ |

**User B Key Generation**

| | |
|---|---|
| Select private $n_B$ | $n_B < n$ |
| Calculate public $P_B$ | $P_B = n_B \times G$ |

**Calculation of Secret Key by User A**

$K = n_A \times P_B$

**Calculation of Secret Key by User B**

$K = n_B \times P_A$

**Figure 10.7 ECC Diffie-Hellman Key Exchange**

# Security of Elliptic Curve Cryptography

- Depends on the difficulty of the elliptic curve logarithm problem
- Fastest known technique is “Pollard rho method” $\rho$
- Compared to factoring, can use much smaller key sizes than with RSA
- For equivalent key lengths computations are roughly equivalent
- Hence, for similar security ECC offers significant computational advantages

# Table 10.3

## Comparable Key Sizes in Terms of Computational Effort for Cryptanalysis (NIST SP-800-57)

| Symmetric key algorithms | Diffie-Hellman, Digital Signature Algorithm | RSA (size of $n$ in bits) | ECC (modulus size in bits) |
|---|---|---|---|
| 80 | $L = 1024$<br>$N = 160$ | 1024 | 160–223 |
| 112 | $L = 2048$<br>$N = 224$ | 2048 | 224–255 |
| 128 | $L = 3072$<br>$N = 256$ | 3072 | 256–383 |
| 192 | $L = 7680$<br>$N = 384$ | 7680 | 384–511 |
| 256 | $L = 15,360$<br>$N = 512$ | 15,360 | 512+ |

*Note:* $L$ = size of public key, $N$ = size of private key

Public key cryptography fundamentally relies on the concept of a trapdoor function. It operates like a sophisticated trap: a rat (representing the data) gets caught using the public key, but it cannot escape on its own. However, with the knowledge of a specific method—the private key—the rat can be freed effortlessly. This elegantly demonstrates the power and precision of mathematics, ensuring that while anyone can set the trap (encrypt), only the intended recipient can unlock it (decrypt) 😊 .

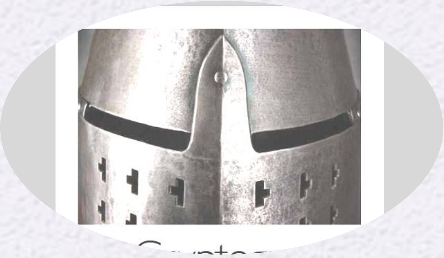
using public key

#PublicKeyCryptography
#Security
#RSA

using private key

# Summary

- Define Diffie-Hellman Key Exchange
- Understand the Man-in-the-middle attack
- Present an overview of the Elgamal cryptographic system



- Understand Elliptic curve arithmetic
- Present an overview of elliptic curve cryptography
- Present two techniques for generating pseudorandom numbers using an asymmetric cipher